04-18 第二篇

How Rainbows Form: Science Behind the Colors

Have you ever seen a rainbow after a rainstorm? The curved band of red, orange, yellow, green, blue, and purple across the sky seems almost magical. [1] In fact, rainbows are not magic—they are a natural result of light and water working together. To see a rainbow, you need three things: sunlight, water droplets（水滴） in the air, and your eyes facing away from the sun. When these conditions are right, a rainbow can appear in the sky or even near a waterfall or garden hose（水管）.

Light from the sun looks white, but it is actually made up of many different colors. [2] When sunlight enters a water droplet, it bends—this is called refraction（折射）. Inside the droplet, the light splits into the colors of the spectrum（光谱）. Each color bends at a slightly different angle, so they separate. Then the light hits the back of the droplet and bounces back toward your eyes. This bounce is called reflection. So refraction and reflection inside each tiny droplet create the colors we see.

Not everyone sees the same rainbow. [3] Each person sees a rainbow from their own viewing position. The rainbow you see is made of light from certain droplets that are at the right angle to reach your eyes. Someone standing next to you sees light from slightly different droplets, so their rainbow is a little different from yours. In a sense, every rainbow is personal—it is created by the light that reaches your eyes and no one else's.

Rainbows always appear in the opposite direction from the sun. [4] If the sun is behind you, the rainbow will form in front of you, often when rain is falling in the distance. The center of the rainbow's arc is always in the "anti-solar point"—the point directly opposite the sun in the sky. You can never walk to the end of a rainbow because it moves as you move. It is like chasing a shadow; the rainbow stays at a fixed distance from your eyes.

[5] All in all, rainbows are a beautiful example of how sunlight, water, and our eyes work together. Next time you see one, remember the science: millions of tiny water droplets are bending and reflecting light, and you are in just the right place to catch that colorful show. Nature is full of wonders that we can understand if we look a little closer.

选项

A. The science behind it involves two main processes: refraction and reflection. B. Rainbows follow a fixed rule related to the position of the sun. C. That is because a rainbow is formed by the light that enters your eyes from specific droplets. D. In other words, rainbows show us that science can explain many natural wonders. E. To sum up,

rainbows are easy to enjoy but understanding them makes the experience even richer. F. You need to have the sun behind you and rain or mist in front of you.

答案键（Answer Key）

[1] D [2] A [3] C [4] B [5] E

深度解析

- **[1] D**：
  - **线索**：从"魔术感"过渡到科学解释。空前描述"rainbow...seems almost magical"，空后指出"rainbows are not magic—they are a natural result of light and water...To see a rainbow, you need three things"。D选项"In other words, rainbows show us that science can explain many natural wonders"（换句话说，彩虹向我们展示科学能解释许多自然奇观）中"In other words"承接"似乎神奇"，"science can explain"与空后"not magic...natural result"精准衔接，引出三要素说明。
- **[2] A**：
  - **线索**：白光与两种物理过程的引入。空前说明"Light from the sun looks white, but it is actually made up of many different colors"，空后详述"sunlight enters a water droplet, it bends—refraction...light splits...reflection...refraction and reflection inside each tiny droplet create the colors"。A选项"The science behind it involves two main processes: refraction and reflection"（其背后的科学涉及两个主要过程：折射和反射）总领空后具体解释，与空后术语及"create the colors"紧密呼应，衔接自然。
- **[3] C**：
  - **线索**：个人视角与不同观察者。空前提出"Not everyone sees the same rainbow...Each person sees a rainbow from their own viewing position"，空后解释"The rainbow you see is made of light from certain droplets...Someone standing next to you sees...slightly different droplets...every rainbow is personal"。C选项"That is because a rainbow is formed by the light that enters your eyes from specific droplets"（这是因为彩虹是由进入你眼睛的、来自特定水滴的光形成的）中"formed by the light that enters your eyes from specific droplets"与空后"light from certain droplets...at the right angle to reach your eyes"精准对应，因果衔接顺畅。
- **[4] B**：
  - **线索**：太阳位置与彩虹方位的规律。空前描述"Rainbows always appear in the opposite direction from the sun...If the sun is behind you, the rainbow will

form in front of you...center of the rainbow's arc...anti-solar point...You can never walk to the end"。B选项"Rainbows follow a fixed rule related to the position of the sun"（彩虹遵循与太阳位置相关的固定规律）总领空后具体规则，与空后"opposite direction...sun behind you...anti-solar point"呼应，逻辑紧密。

- **[5] E**：
  - **线索**：全文收束与升华。空[5]位于结尾段段首，空后总结"beautiful example of how sunlight, water, and our eyes work together...remember the science...Nature is full of wonders that we can understand"。E选项"To sum up, rainbows are easy to enjoy but understanding them makes the experience even richer"（总之，彩虹容易欣赏，但理解它们会让体验更丰富）用"To sum up"总结全文，"understanding them makes the experience even richer"与空后"remember the science...understand"完美呼应，收束得当。
- **干扰项分析**：
  - **F. You need to have the sun behind you and rain or mist in front of you.**：描述观看彩虹的条件，与空[1]或空[4]相关，易被误选。但空[1]后已列出"three things: sunlight, water droplets, and your eyes facing away from the sun"，F仅重复其中部分内容且过于具体，缺乏总领或过渡功能；空[4]需总领句，空后已直接展开具体规则，F属细节复述。E的"To sum up"为典型总结标记，与结尾段功能匹配。F属"信息正确但位置错位"型干扰。